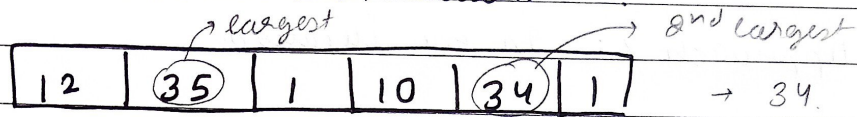


GfG_160_DSA

Day-001

P1: Second largest Element

Problem statement: Given array arr, return the second largest distinct element from an array. If the second largest element doesn't exist return -1.



Sample Test cases:

Input:

12	35	1	10	34	1
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Output: 34

Input:

10	5	10
----	---	----

Output: 5

constraints

$$2 \leq \text{arr.size} \leq 10^5$$

$$1 \leq \text{arr}[i] \leq 10^5$$

Input:

10	10	10
----	----	----

Output: -1

Approach 1: (using sorting)

→

12	35	1	10	34	1	35
----	----	---	----	----	---	----

 ←after sorting:

1	1	10	12	34	35	35
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↑ ↑ ↑

→ return the unique 2nd last element

complexity analysis:

Time complexity: $O(n \log n)$ ($n \log n + n$)Space complexity: $O(1)$

Approach 2: In two iterations

12	35	1	10	34	1	35
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→ find max in 1st iteration

→ find max value (less than the previous) in 2nd iteration.

complexity analysis:

$$T.C : O(2n) \approx O(n)$$

$$S.C : O(1)$$

Approach 3: In one iteration.

→ look for max & keep track of the previous updated value & 2nd max value.

→	12	35	1	10	34	1	35	sec	max
	↑	↑	↑	↑	↑	↑	↑	12	35
					=			34	12
									35

complexity analysis:

$$T.C. \Rightarrow O(n)$$

$$S.C \Rightarrow O(1)$$